Editor-in-Chief
Journal of Information and Communication Technology
Universiti Utara Malaysia Press

Dear Editor,

We submit the manuscript "When the Lab Remembers: Identifiability and budgeted confirmation of materials recipes" for consideration as an original research article in the Journal of Information and Communication Technology.

The work falls within the journal's stated scope: A computational methods and failure-analysis study of confirmation under experimental carryover, with a reanalysis of published measurements; we note that its subject, materials informatics for autonomous laboratories, sits at the edge of the journal's information and communication technology scope and we leave that judgement to the editor.

The paper studies confirmation of one frozen materials recipe under independent fresh preparation when residual material in shared equipment makes the measured preparation differ from the recorded recipe. A finite-data analysis distinguishes underdetermination without quantitative regularity from sharp response envelopes under a known Lipschitz bound; a mass-balance memory model yields a contraction-based preparation-bias bound and a conditional fixed-final confidence interval; an exchange argument establishes stronger-wash-first ordering for fixed action counts. A reanalysis of original organic-photovoltaic measurements finds 10 threshold disagreements among 240 paired formulations without establishing carryover causality. A preserved synthetic benchmark of 48,000 policy runs does not demonstrate superiority of an adaptive heuristic over same-model repetition, a negative finding kept in full. A separate extension of 62,400 runs shows the planner improving true-acceptance yield over costly fresh-only measurement by 3.5 and 4.0 percentage points in high-carryover cases but losing 3.0 points to fixed burn-in/tail repetition when washing is ineffective; persistent-memory stress produces 12 false acceptances in 200 campaigns for the misspecified adaptive model and observed coverage falls below nominal in some controls. The emulators' certificates do not certify physical materials and the paper says so. For the journal's 8,000-word limit the submitted text keeps every result, limitation and citation; the full appendix derivations are in the master manuscript held in the repository named below.

The manuscript is not under consideration elsewhere and has not been published previously. No part of it has been presented at a conference. The study code, the frozen protocol, the scored results and the figures are in a repository that is private at the time of submission and will be made public on request of the editors or reviewers and on acceptance: https://github.com/bdata-arindam/labremembers-paper. Because this journal uses double-blind review, the anonymous manuscript states availability without naming the repository; the address is given here and in the complete version.

No funding was received for conducting this study. The authors have no competing interests to declare that are relevant to the content of this article. No institutional determination applies to this study. All four authors have approved the submission. The manuscript names three corresponding authors, Arindam Chakraborty, Sambit Dhar and Soham Marik; correspondence on this submission is handled by Arindam Chakraborty, the first author.

Yours sincerely,

Arindam Chakraborty
Independent Researcher, Leander, TX, United States
arindamc.81@ieee.org
on behalf of Arindam Chakraborty, Sambit Dhar, Soham Marik and Surajit Sinha